

Design Solution	Description
“Portal”	The design allows the daycare members to attach the wristband to the child’s wrist using only one hand. This allows the members to potentially multitask and perform other morning duties as necessary. The design allows the wristband to be placed into a circular “C” shaped structure, with the “male” part of the snap sitting in a divot in the tab. This allows the wristband to stay in position, making the application process easier. The tab also acts as a tool for ensuring the wristband is not too tight. Then, with the wrist still inside the structure, the staff member can cut off any excess as required. The wrist can then be removed and the wristband will be attached.
“Magnetomatic”	The design mainly contains a bracelet with two magnetic sides and a metal skeleton. The two magnets are attached to the skeleton before being applied. The user needs to put one of the hands between the bracelet and the skeleton first, and move his or her hand toward the bracelet. By exerting force to the bracelet, the magnet on the top will detach from the skeleton and fall down due to gravity. It can automatically attach to the magnet at the bottom and therefore wrap the wrist. Once the user keeps moving the hand apart, the second magnet will detach from the skeleton.
“Knobby”	This design takes the approach of creating a new wristband that better fulfills the stakeholder’s needs. The band is made from a sturdy but comfortable material that is cognizant of children’s sensory needs as a primary stakeholder. The ends of the wristband meet at a round mechanism, not larger than the size of a watch. This is also made of a safe material that is comfortable at where it's in contact with the skin. When rotated clockwise, this mechanism tightens the loop part of the bracelet. If squeezed and rotates counter clockwise, it “pops out” and allows the user to loosen the loop. This acts as a sturdy and durable child lock that also has space for an identification mechanism on top.

Design Solution	Advantages	Disadvantages	Requirements
Portal	1. Five-step process 2. Speed - quick to use, minimal set-up required 3. Adjustable for all wrist sizes 4. Minimal force is applied to the user's wrist	1. Has a potential pinching point (the clasp), making it unsafe for children 2. Potentially difficult to use and hard to use for people with limited fine motor skills	All requirements met
Magentomatic	1. Requires less than 5 steps to complete, the whole process is less than 34 seconds 2. Adjustable to all wrist sizes 3. Reusable and durable 4. Force applied to the user's wrist is adjustable so can be minimized	1. Magnets are heavy, making the wristband heavy 2. Difficulties for young kids to wear wristbands by themselves 3. The process when the magnet fall down might hit the wrist or hand of the kid, might cause potential safety issues	All requirements met
Knobby	1. Only quick, simple actions needed to apply 2. Easily adjustable to different children 3. Can be removed and reused	1. Still may require staff assistance and energy 2. Potential for temporary pinching if applied hastily 3. Relatively large (size of children's analog watch) excess material - may be uncomfortable	All requirements met

Evaluation Criteria	Potal	Magentomatic	Knobby
Set up time (seconds): shorter is better			
Number of steps: less is better			
Application time per wristband (seconds): shorter is better			
Total preparation time (minutes): shorter is better			
Calculate the verification time per child. The time has to be ≤ 5 seconds. Shorter time is better.			
Record the time spent before the wristband is not usable (minutes): Shorter time is better.			
Complexity of actions per step. Simpler actions are better			
Learning time for process. Shorter is better			
Percentage of users accommodated. Higher is better			
Wrist clearance. Adequate clearance without looseness			
Force before rupture. Higher is better.			

Deformation of the tape. Lower is better			
Ratio of rounded edges to sharp corners. Higher is better.			
Gap sizes of <5mm, or >15mm, the better.			
Speed of moving parts (m/s). Lower is better.			
Force withstood. Higher is better.			
Cycles of testing without failure. Higher is better.			
Force applied to wrist. Lower is better.			
Distribution of force. Higher is better.			
Force required by wrist. Lower is better.			
Duration of applied force. Lower is better			
Final Count	12  7  2 	12  7  2 	14  5  2 
Legend  Satisfies well  Satisfies ok  Satisfies poorly			